Natural Language Processing

Natural language processing, or NLP, is a field of artificial intelligence that enables computers to analyze, understand, and generate human language. It is used in chatbots, translation tools, search engines, voice assistants, text summarizers, and spam filters. Language is challenging because the meaning of a word can depend on context, tone, grammar, and cultural knowledge. NLP systems therefore transform text into numerical representations that machine-learning models can process. Some systems classify text into categories, while others generate new sentences or answer questions. Recent language models can produce fluent responses by predicting likely sequences of words from patterns learned during training. Even so, fluent language does not guarantee factual accuracy or sound reasoning. Users should verify important outputs, especially in education, health, finance, law, and other situations where mistakes could have serious consequences.